

Building and Tuning a AM Radio Receiver

INTRODUCTION

AM radio receivers rely on tuned LC circuits and diodes to isolate and decode amplitude modulated signals from the air. In this lab, we explore how to detect and demodulate AM radio waves using inductors, capacitors, diode and Antenna. The goal is to construct a radio circuit that can pick up real radio broadcasts and convert them into audible sound through a speaker.

THEORY

(~) The resonant frequency ω_0 of an LC circuit is given by:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

(~) In terms of frequency f in Hertz:

$$f = \frac{1}{2\pi\sqrt{LC}}$$

(~) The inductance L of a solenoid with N turns, diameter D , and wire diameter d is:

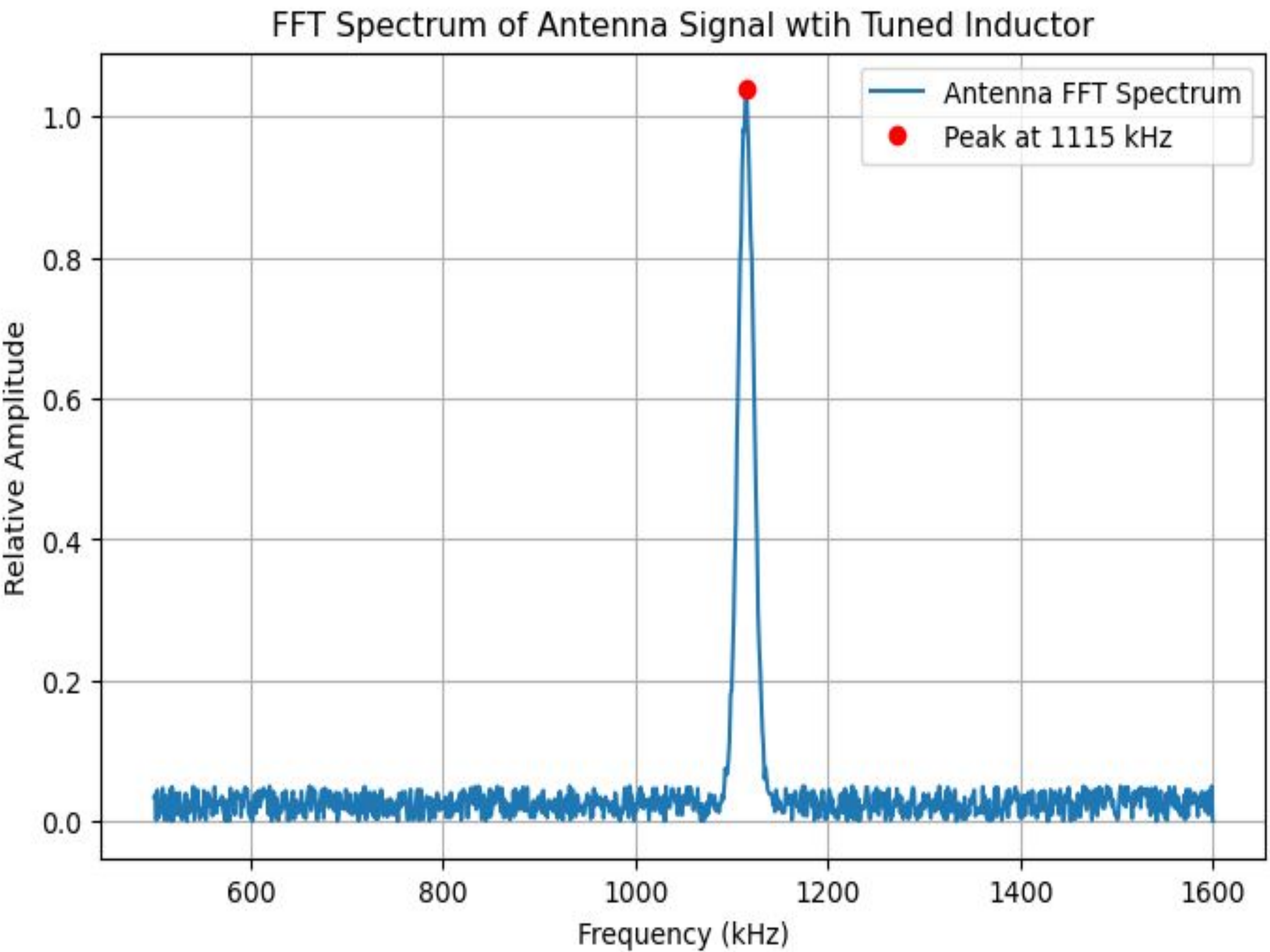
$$L = \frac{\mu_0 \pi N D^2}{4d}$$

(~) This resonant frequency allows the circuit to filter out a specific AM radio signal from the wide range of frequencies picked up by the antenna. The diode then demodulates the amplitude of the carrier wave, converting it into an audio signal. The final signal can be observed with an oscilloscope and heard through a speaker.

METHODS

Using the oscilloscope, the FFT of the antenna signal was analyzed to identify peaks corresponding to local radio stations. Based on the target frequency, a resonant LC circuit was designed using a homemade inductor and a variable capacitor (40–220 pF). The inductance was verified with an LCR meter. A small pick-up coil was wrapped near the antenna to couple the signal inductively into the LC circuit. A diode was then used to demodulate the signal, and a speaker was connected to convert the output into sound. The frequency response and station tuning were observed using the oscilloscope FFT display.

Findings and Results

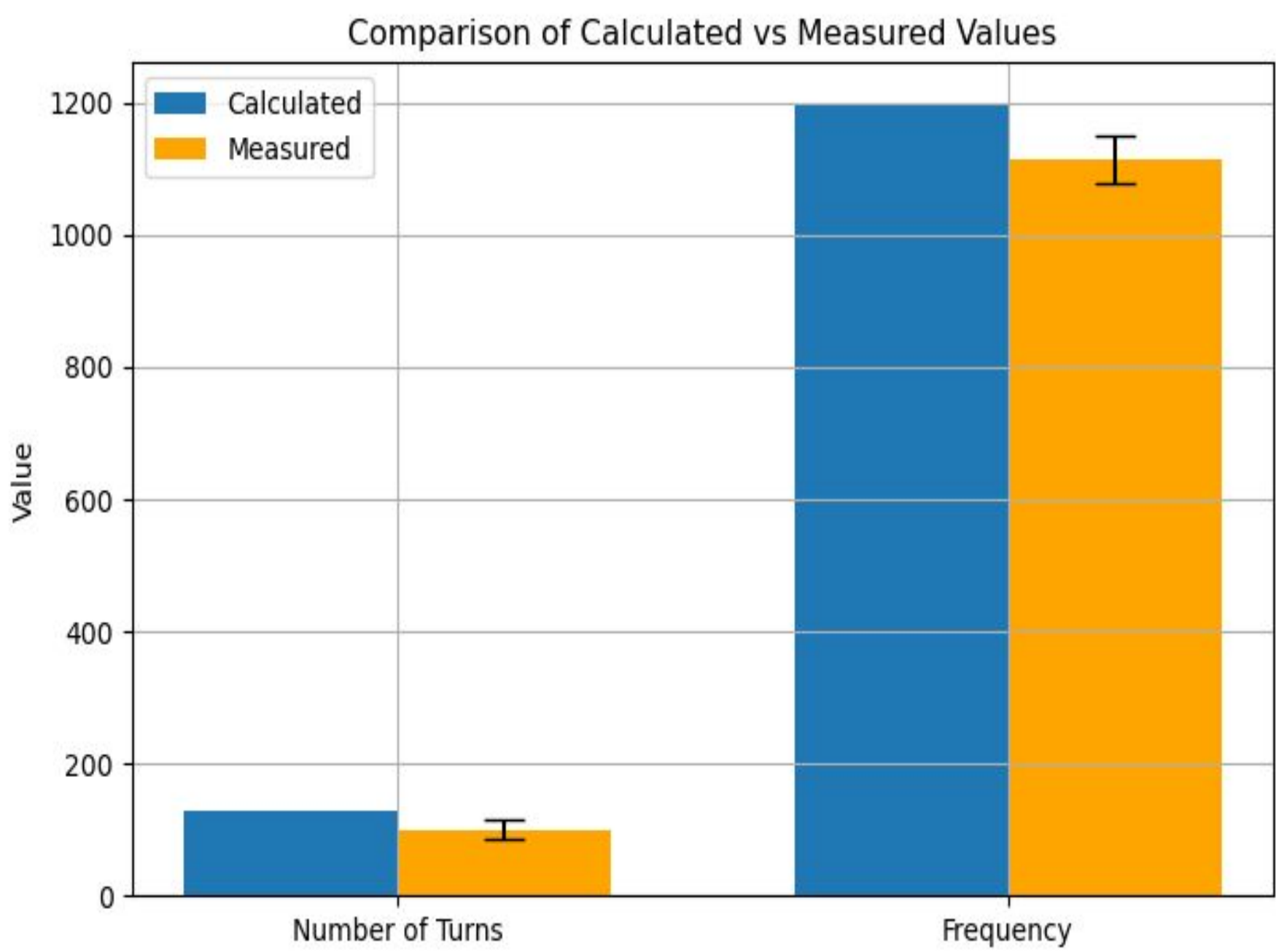


The plot shows that our homemade radio successfully captured a real AM station broadcasting at 1200 kHz. The small random noise across the plot represents weak background signals and electrical noise. They are not real data but rather simulated using a uniform random exponential distribution.

Dominant Source of Uncertainty

The dominant source of uncertainty in this lab was how the wire was wound to form the inductor. Ideally, for a uniform magnetic field, the wire turns should be tightly and evenly spaced like a true solenoid. However, in practice, the loops were not closely packed, and the spacing varied throughout the coil. This irregular winding caused the magnetic field inside the coil to be non-uniform, leading to deviations from the expected inductance value. As a result, the actual resonant frequency differed significantly from the calculated target and the sound heard from the speaker was falling.

Comparison To Accepted Value

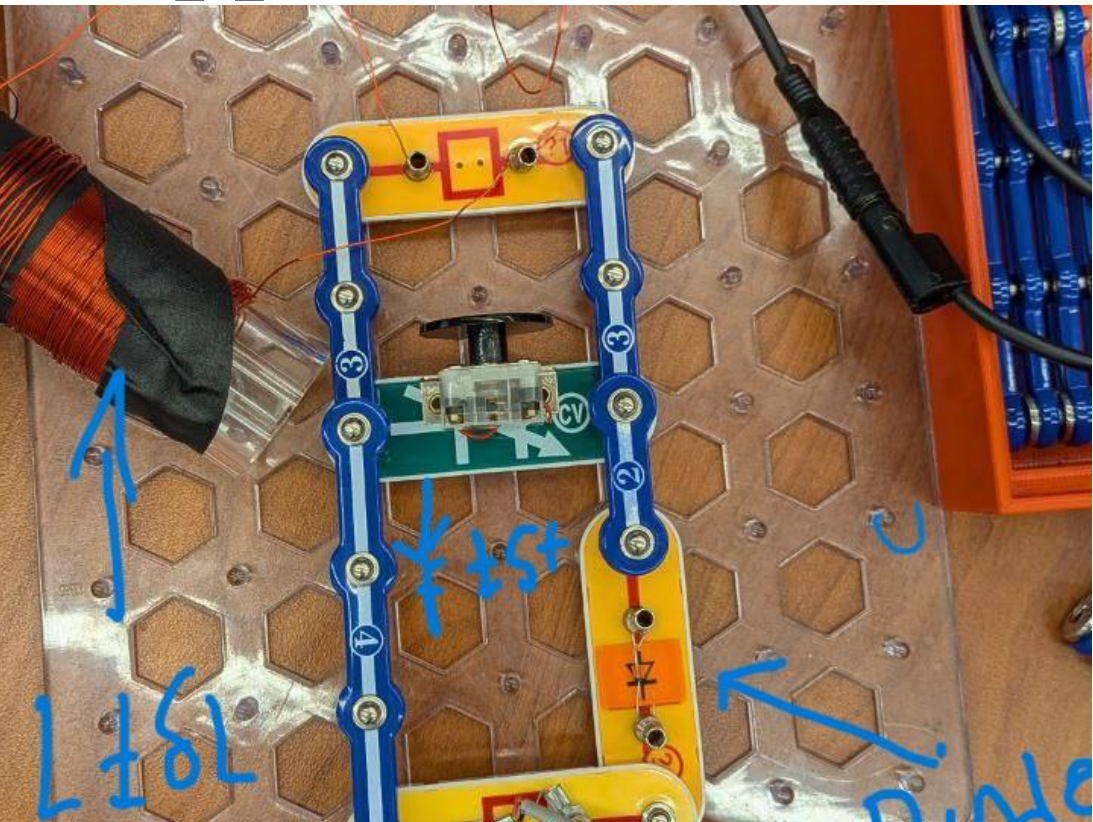


Experimental Reflection

The initial coil had loose and uneven loops, which made the magnetic field inside the coil non uniform and lowered the inductance. This shifted the resonant frequency below the target. After several iterations of re looping the wire and making the turns closer together, the magnetic field became more uniform and the inductance increased toward the calculated value. The resonant peak got closer to the desired frequency, and eventually allowed the circuit to tune correctly and pick up a real station around 1115 kHz.

It was surprising how much small changes in coil spacing affected the resonance. The small physical imperfections made it difficult to listen to a clear audible voice from the frequency.

Apparatus



Analysis PDF

